

Cloud computing fundamentals (Discover how cloud computing helps business move faster, do more, and scale with less effort) :-

Imagine you have a fantastic idea for an app that helps students practice Cyber Security, you host it on your computer in your country. But what can you do when users from other parts of the world try to access it and experience lag? What if many students connect at the same time, or your computer is turned off? These limits make it hard for an app to grow. That's where cloud computing comes to play and solves these problems!

The cloud is built on top of technologies you already learned, like virtualization and containers. These help running many applications efficiently on shared infrastructure and quickly creating or changing environments when needed.

In this room, we will explore basics of cloud computing and how it impacts the way modern applications are built, deployed, and used every day.

Learning objectives :-

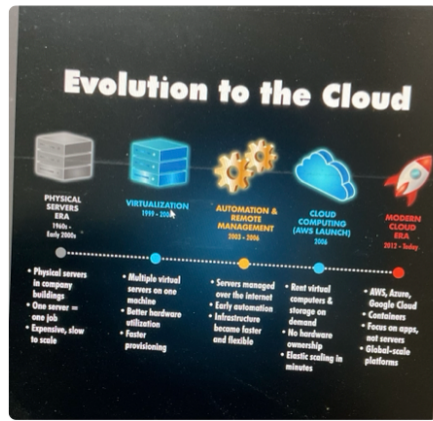
- 1) What is cloud computing
- 2) Service models of cloud (IaaS, PaaS, SaaS)
- 3) Cloud types (private | public | hybrid)
- 4) Benefits of cloud computing
- 5) How big companies are using cloud.

Task 1 :- cloud computing overview

Cloud computing is a perfect solution for challenges in application you're designing, such as moving your files from a single laptop to online storage that you can access anywhere. Instead of running your app on one country, cloud lets you use computing resources over the Internet. This makes application easier to access, more reliable, and ready to grow as more computers start using it.

How servers evolved to cloud :-

Before we start diving into cloud details, it's helpful to understand that cloud computing didn't appear suddenly. It is the result of many years of changes in how servers were used and managed. At first, businesses looked for ways to reduce costs, use resources more efficiently, and make their applications easier to run and scale. The timeline below shows the evolution from physical servers to cloud as we use



Cloud Benefits and Characteristics :-

After seeing how Applications evolved from physical Servers to cloud, it becomes clear why Cloud Computing is so widely used today. The cloud was designed to address common problems, including limited capacity, high costs and slow growth.

The following benefits and characteristics explain how cloud computing makes applications easier to run, scale and manage :-

- * **Scalability** :- Easily scale up or down as your application's need change.
- * **On demand Self service** :- Create or remove servers and storage instantly, without waiting for hardware.
- * **Pay only for what you use** :- you are charged based on usage, not upfront costs.
- * **Security** :- Cloud provides infrastructure with strong security measures.
- * **High Availability** :- Applications keep running even if part of system fails.
- * **Global Access** :- Your Application can be accessed by users anywhere in the world.

In simple terms, cloud enables IT resources to be flexible, cost effective and easier to manage.

Types of cloud :-

The flexibility provided by cloud computing allows applications to run in different ways, depending on your needs and level of control. Because of this, cloud providers offer multiple models for deploying and using applications, each suited to different scenarios.

Let's start with deployment types you can choose for cloud environment :

- * **Public Cloud** :- used by startups, websites, and global apps because it's affordable and easy.

to Scale, and requires no Infrastructure management. Public cloud services are preferable for every use case.

* **Private cloud**: — used by banks, healthcare and government organization because it offers great control, customization and compliance for sensitive data.

* **Hybrid cloud**: — used by companies like e-commerce platforms that needs to keep sensitive data private, while scaling publicly during high demand.

Just like they are different ways to deploy cloud environment, they are also different ways to use cloud services. Depending on your experience and needs, you can choose level of responsibility that fits your application. looks like there are main cloud service models.

* **Infrastructure as a Service (IaaS)**: —

you rent basic computing resources such as virtual servers, storage and networking. you are responsible for managing operating system and your application, while provider manages physical hardware.

* **Platform as a Service (PaaS)**: — The cloud provider manages infrastructure and operating system. You focus on building, deploying and running your application without worrying about servers.

* **Software as a Service (SaaS)**: — you use a complete application over the Internet. The provider manages everything, and you access software through browser or app, for example Gmail or Zoom.

IaaS Examples are AWS EC2, Microsoft Azure virtual machines, GCE (Google compute engine).

PaaS Examples are AWS Elastic Beanstalk, Heroku, Google App Engine.

Major cloud vendors: —

AWS is industry leader, with most extensive offerings and global reach. other well known cloud providers include: —

* **Microsoft Azure**: — A strong competitor, especially in enterprise and hybrid cloud environments.

* **Google cloud platform (GCP)**: — Known for powerful data analytics, AI and ML tools.

* **Alibaba cloud**. * **IBM cloud** and **Oracle cloud**

How companies are using cloud: -

- * Netflix runs on AWS platform, stay online during peak demand, and stream content at once.
- * Spotify uses cloud to handle millions of songs and users, scaling quickly new music.
- * Instagram uses cloud to store massive amount of photos and videos and deliver fast.

Quiz:-

- | | |
|--|---|
| 1) What is characteristic of cloud environment that enables you to handle and unexpected increase in users to your application? | 2) What is most common type of cloud deployment used? |
| A) Scalability | A) Public cloud |
| 3) Suppose you want to deploy an application to Internet, focusing only on application development and leaving infrastructure to others. What type of cloud service is best? | |
| A) PaaS | |